Rockborne

Intro to Data Science

What is Data Science?

Objectives

How DS is Different

vs engineering and analytics.

Main Tools

The data scientist's toolkit.

Normal Workflow

The data science process.

Machine Learning

What it is and its categories.

Learning Path

For a new data scientist.

Intro to Data Science

How Data Science is Different

From Engineering or Analytics?

What is Data Science?

The famous definition

- "A person who is better at statistics than any software engineer, and better at software engineering than any statistician."
- Josh Wills, former Director of Data Science, Cloudera

In short

- Blends statistics, programming and domain knowledge.
- Extracts insight and predictions from data.
- Sits between engineering and analytics.

Source: iSchool Berkeley - What is Data Science?

How Data Science is Different

Responsibilities

Core Skills

Programming

Python, R and clean, reproducible code.

Statistics

Inference, testing and uncertainty.

Machine Learning

Models that learn from data.

Data Visualisation

Communicating findings clearly.

Data Science vs Data Engineering

Data Science

- Gathering and processing data.
- Machine learning and statistics.
- Exploratory data analysis.
- Visualisation and analytics.

Data Engineering

- Systems building.
- Methodology oriented.
- Programming and database services.
- Coding in multiple languages.

Source: Edureka - Analyst vs Engineer vs Scientist

What Makes Up a Data Science Team?

Product managers

Own the problem and priorities.

Software engineers

Build the surrounding product.

Data engineers

Pipelines and data access.

Operational experts

Process and workflows.

Model integrators

Put models into operations.

Onboarding & training

Help people adopt the solution.

End users

Ultimately use the solution.

Value quantifiers

Measure the value generated.

Intro to Data Science

Data Scientist Tools

The Data Scientist Tool Set

Python / R

The core languages.

SQL & databases

Storing and querying data.

Data visualisation

matplotlib, seaborn, BI tools.

ML libraries

scikit-learn and friends.

Statistics

The reasoning underneath.

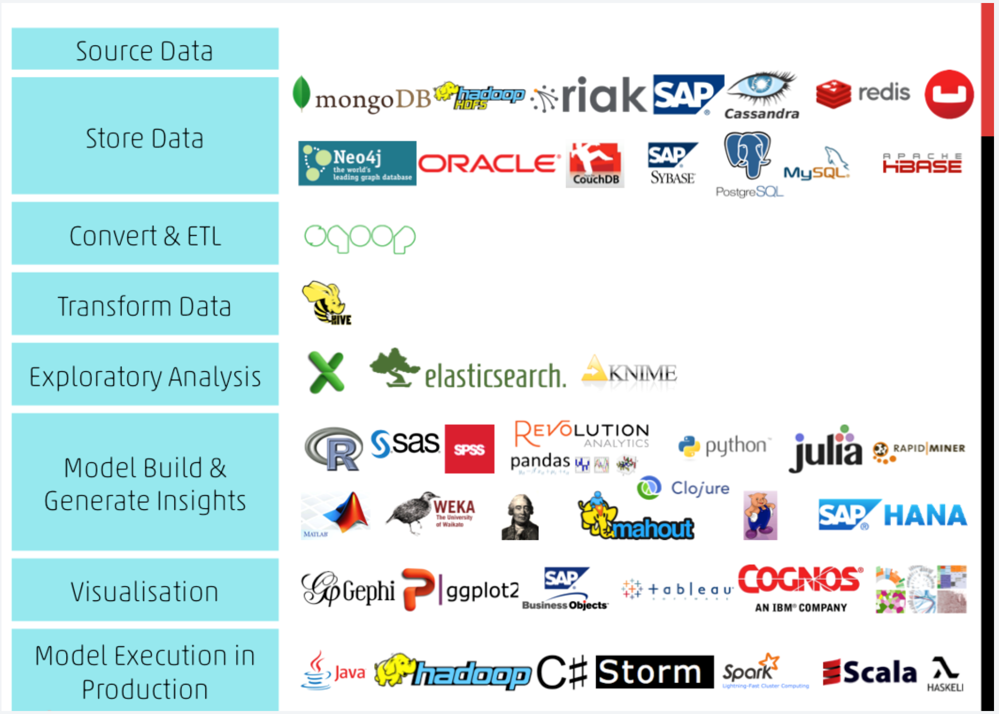
Big data tools

Spark and the cloud.

[Source: Analytics Vidhya - Essential tools for a data scientist]

Tool Sets In Depth

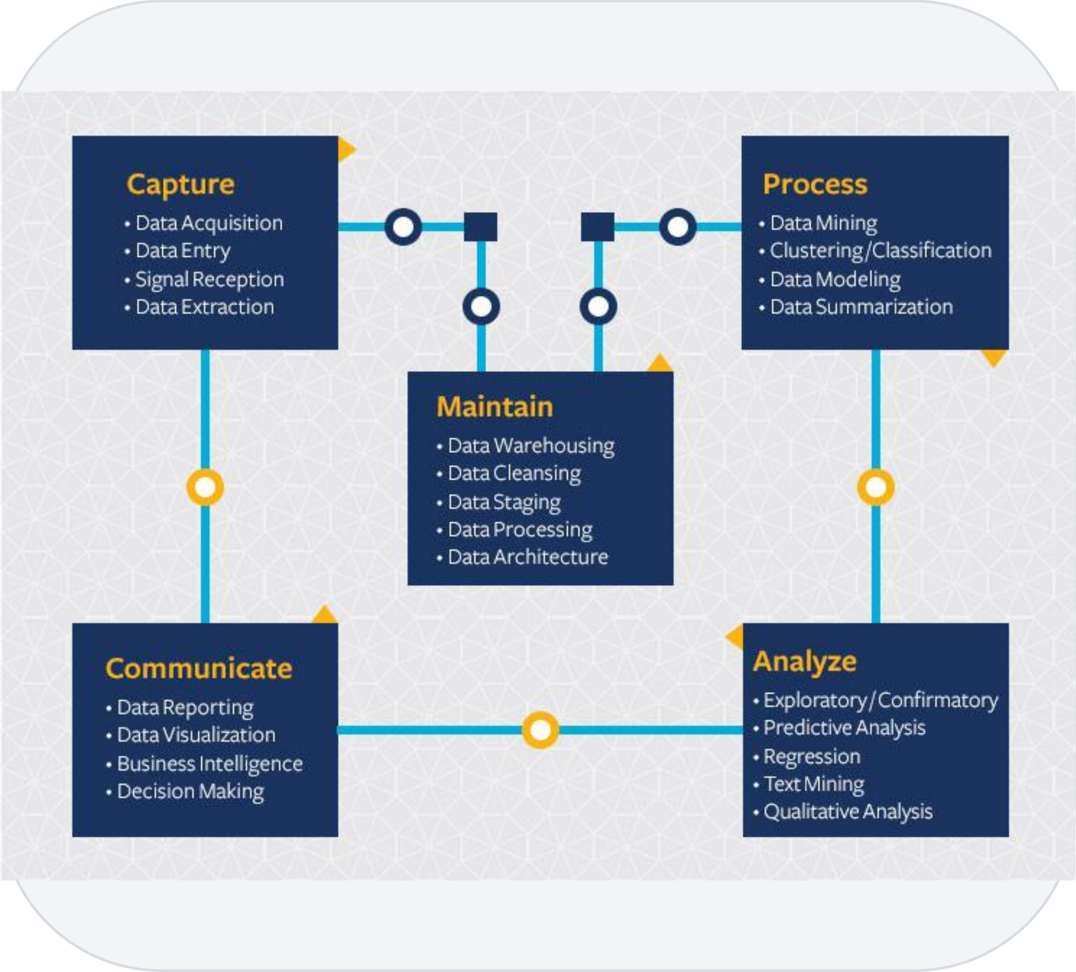
- Every stage of the workflow has its own tools.
- Store and query data; connect and ETL.
- Transform data, then model and generate insights.
- Visualise findings and deploy models to production.



Intro to Data Science

Data Scientist Workflow

The Data Scientist Workflow



1. Capture: acquire and ingest the raw data.
2. Process and maintain: clean, store and stage it.
3. Analyse: explore, model and mine for insight.
4. Communicate: visualise and report the results.

Intro to Data Science

Machine Learning

What is Machine Learning?

The idea

The science and art of programming computers to learn from data.

Learns, not told

It finds patterns instead of following hand-written rules.

Arthur Samuel, 1959

'The ability to learn without being explicitly programmed.'

Source: Accounting Today - What's the big deal about ML?

Machine Learning (via Forbes)

A subset of AI

ML is one branch of artificial intelligence.

Two key steps

Collect and train the data, then operationalise it.

Learned automatically

The model is built from training data, not coded by hand.

Only as good as the data

It needs fresh, updated data for accurate predictions.

[Source: Forbes - Machine learning: what is it really good for?]

Intro to Data Science

Learning Path for a New Data Scientist

Sequence of Recommended Learning

1. Programming (Python)
2. Statistics
3. Data handling & SQL
4. Machine learning
5. Projects & deployment

Learning Path for a New Data Scientist

The Data Science Workflow

What is Data Science? Three Definitions

IBM (2019)

Using algorithms, methods and systems to extract knowledge and insight from structured and unstructured data.

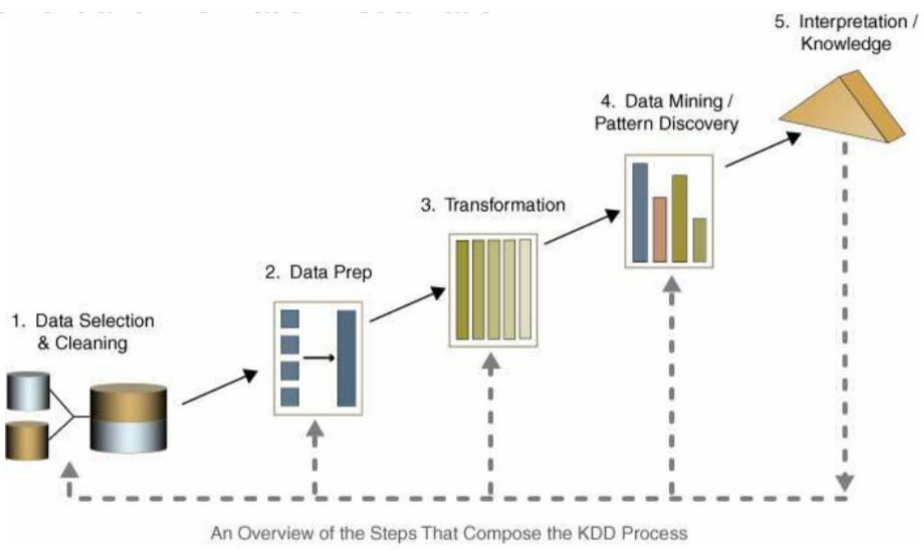
Oracle (2020)

Combines statistics, scientific methods and data analysis to extract value from data.

Hemant Sharma (2020)

A blend of tools, algorithms and ML principles to discover hidden patterns in raw data.

The Data Science Process - KDD



KDD: Knowledge Discovery in Databases.

Selection, then pre-processing of the data.

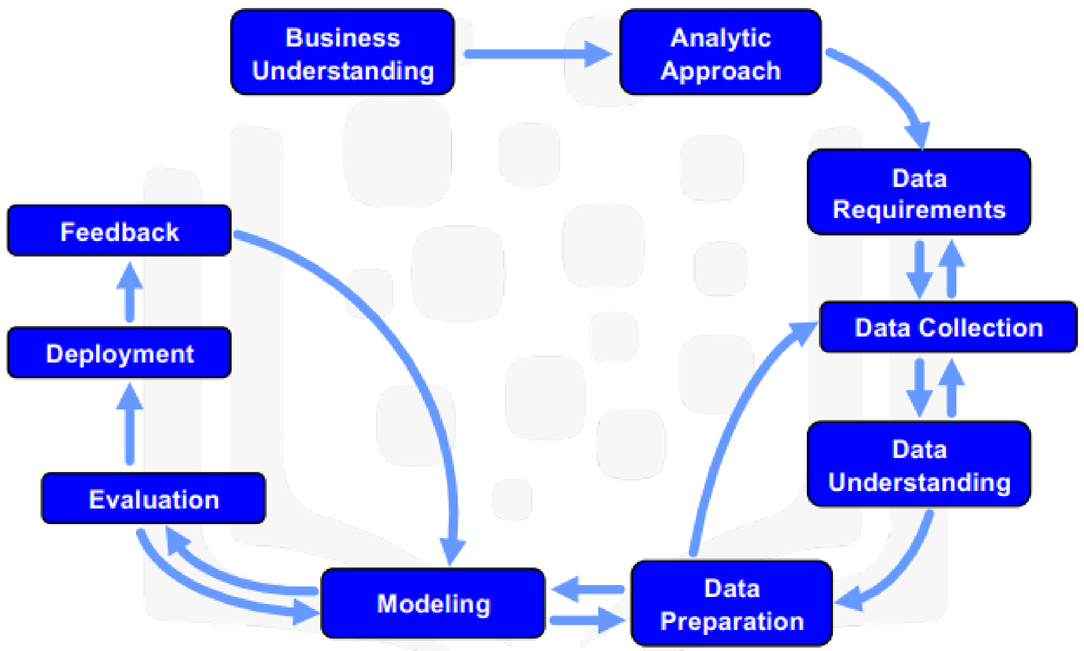
Transformation, then data mining.

Interpretation and evaluation of the patterns found.

[Source: GeeksforGeeks - KDD process in data mining]

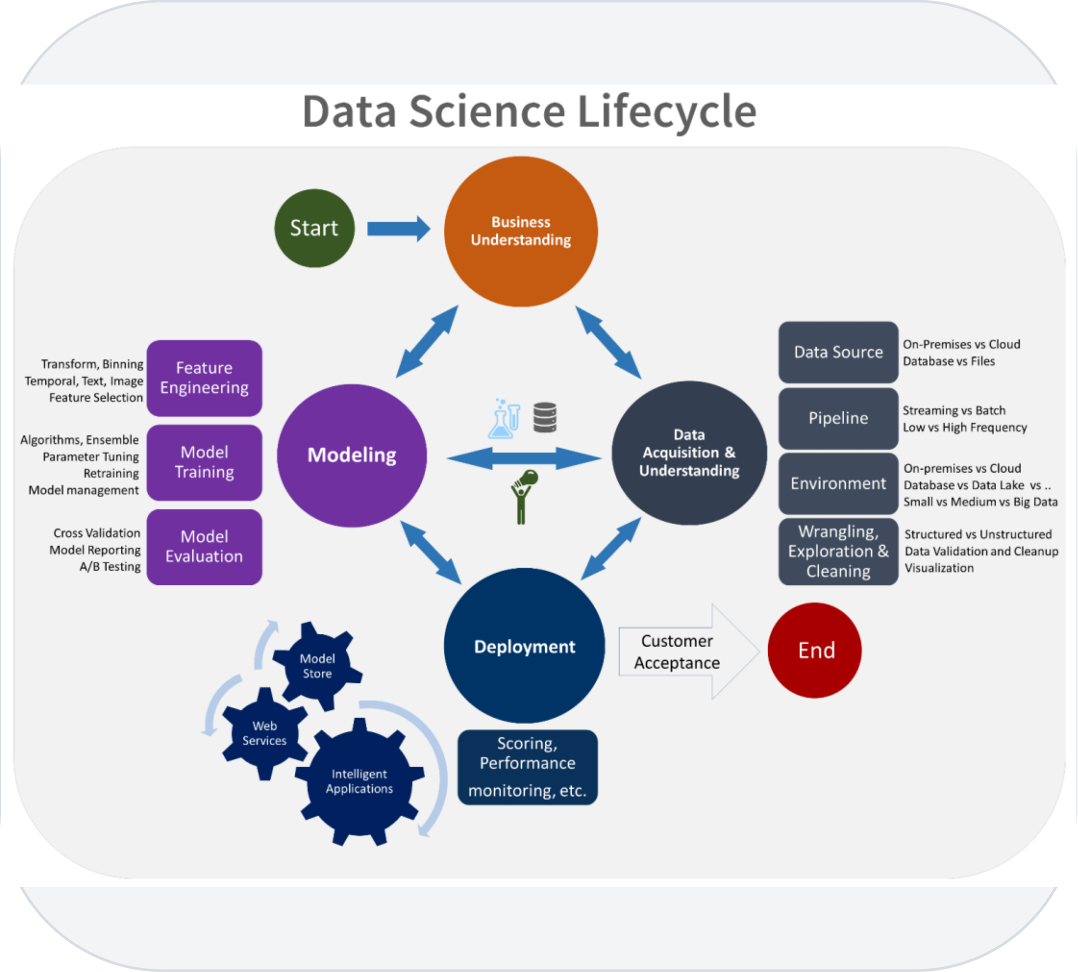
The Data Science Process - IBM

- A business-first, cyclical framework.
- Frame the business and analytic problem.
- Prepare, explore and model the data.
- Evaluate and deploy, then feed back.



Source: IBM - Polong Lin, Foundational Methodology for Data Science

The Data Science Process - Microsoft



Source: Microsoft - Team Data Science Process overview

1. The Team Data Science Process (TDSP) lifecycle.
2. Business understanding and data acquisition.
3. Modelling and evaluation.
4. Deployment and customer acceptance - then iterate.

The Data Science Process

1. Understand the problem
2. Gather & access data
3. Prepare & clean
4. Explore & analyse
5. Model, evaluate & deploy

Data Gathering and Understanding

The problem

What problem are we solving?

The objective

What is the goal of the project?

Success metric

How will we measure success?

Data shape

Distributions and data types.

Data quality

How trustworthy is the data?

Initial strategy

Pre-processing, algorithms, test plan.

Data Science Workflow

Data Preparation / Pre-processing

Data Quality: Common Anomalies

Noise

Modification or distortion of original values.

Outliers

Objects very different from most of the data.

Duplicated values

Records that appear more than once.

Missing values

Data not collected or not applicable.

[Source: Informatica - Foundational components of data quality]

Handling Missing Values

Eliminate

Drop the affected records.

Ignore

Leave them out during the analysis.

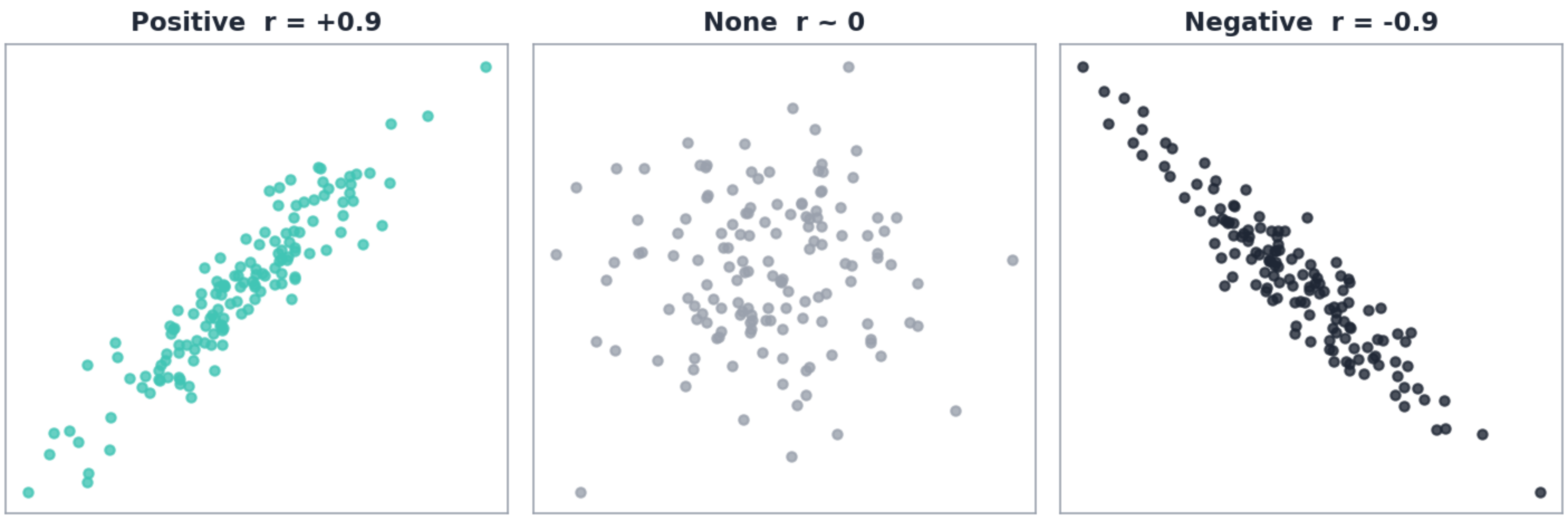
Estimate

Fill with a default, the mean/median, or the most common value.

Data Preparation

Data Reduction and Transformation

Correlation



Pearson's coefficient measures how two variables move together.

It ranges from -1 to +1.

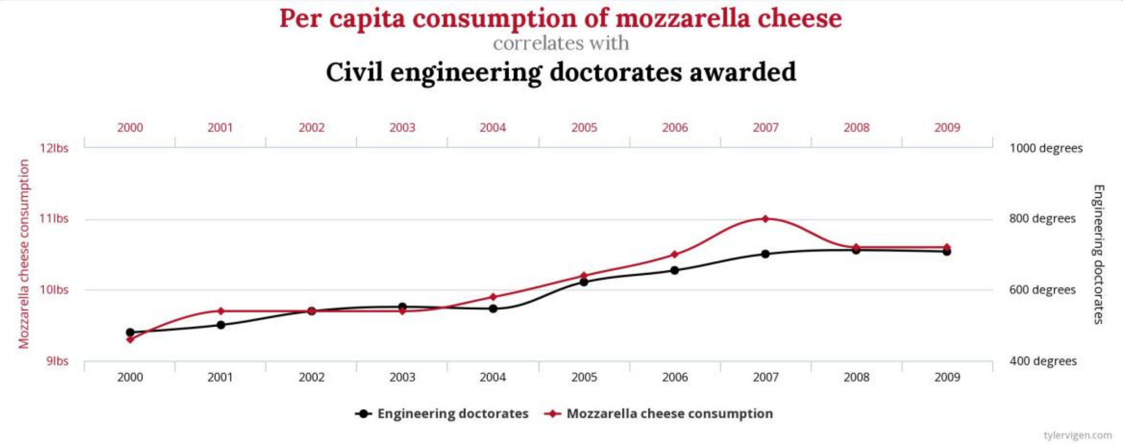
Positive: they rise together; negative: one rises as the other falls.

A value of +/-1 means a perfect linear relationship.

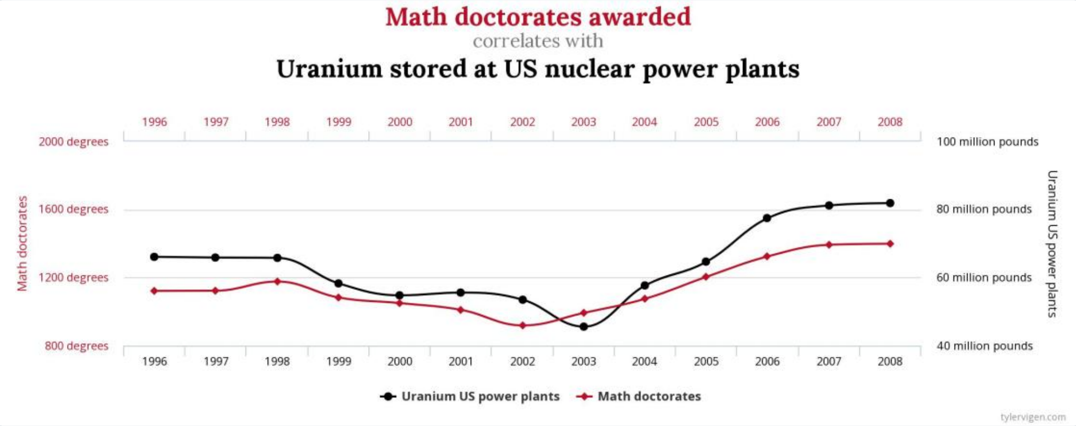
[Source: GeeksforGeeks - Exploring correlation in Python]

Correlation vs Causation

- Causation implies correlation, but not the other way around.
- Correlated variables need not be causally linked.
- With enough tests, spurious correlations appear by chance.
- The chart shows two unrelated series that track each other.



Correlation vs Causation



Source: Tyler Vigen - Spurious Correlations

1. A correlation can occur purely by chance.
2. Or a hidden third variable may drive both.
3. Always ask 'what could explain this?' before claiming cause.
4. More fun examples: tylervigen.com/spurious-correlations.

Transformation: Numerical Data

Why scale?

Different scales bias many algorithms and slow training.

Min-Max

Rescale each feature to a fixed range, e.g. 0 to 1.

Z-score

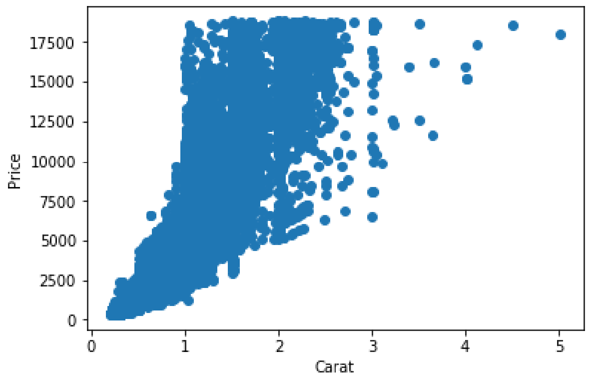
Centre on the mean and scale by the standard deviation.

Normalisation: Min-Max

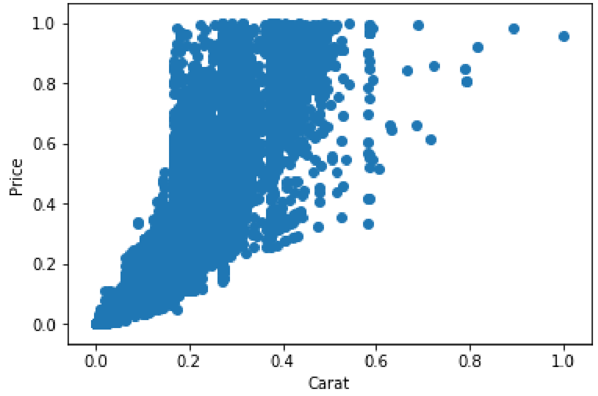
- The minimum value maps to 0 and the maximum to 1.
- The rest of the data is scaled between them.
- One of the most common normalisation methods.
- Downside: it does not handle outliers well.

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Before



After



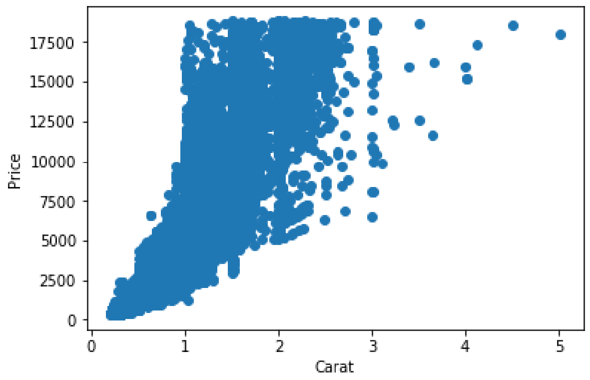
[Source: Codecademy - Normalization]

Normalisation: Z-Score

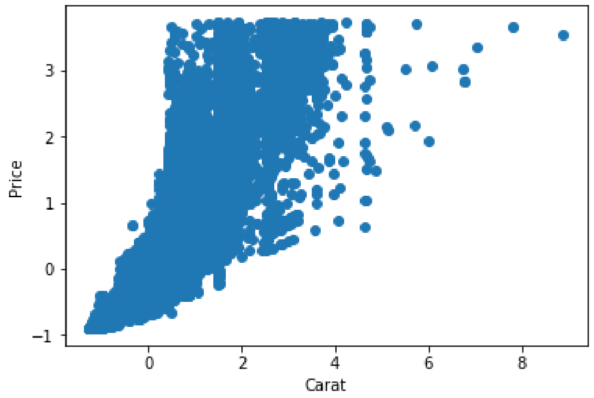
- Subtract the mean, then divide by the standard deviation.
- Values at the mean map to 0; above is positive, below negative.
- A robust default for many models.
- Downside: features may not end on the exact same range.

$$z = \frac{x - \mu}{\sigma}$$

Before



After



[Source: Codecademy - Normalization]

Transformation: Categorical Data

Used directly

Some algorithms (like decision trees) handle categories as-is.

Needs encoding

Most algorithms need categories turned into numbers.

Three methods

Ordinal, one-hot and dummy variable encoding.

Categorical Encodings

Ordinal

An integer per label; for ordered values (first, second, third).

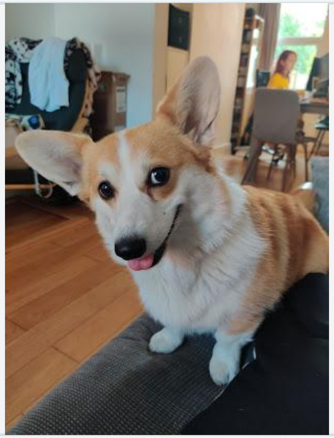
One-Hot

A binary column per label; for unordered values (dog, cat, lizard).

Dummy

C categories as C-1 binary columns; removes redundancy.

Categorical Encoding: Example (Species)



Dog

Ordinal: 1

One-hot: [1, 0, 0]

Dummy: [1, 0]

Cat

Ordinal: 2

One-hot: [0, 1, 0]

Dummy: [0, 1]

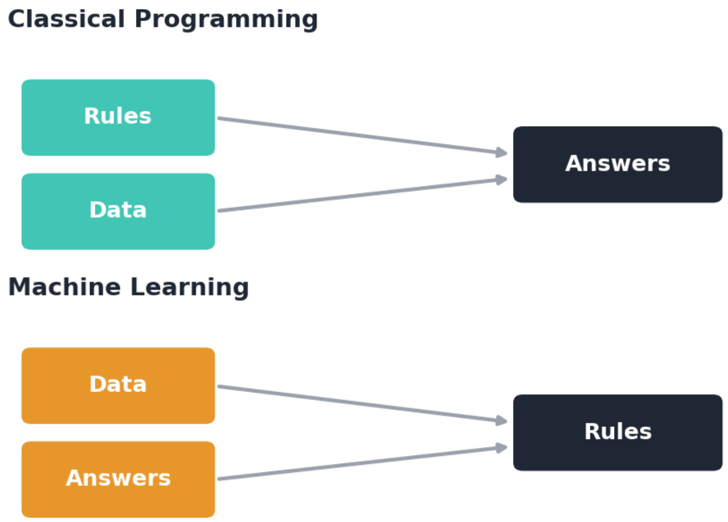
Lizard

Ordinal: 3

One-hot: [0, 0, 1]

Dummy: [0, 0]

What is Machine Learning?



ML learns patterns from data instead of following explicit rules.

'The ability to learn without being explicitly programmed' - Arthur Samuel.

Classical programming: rules + data -> answers.

Machine learning: data + answers -> rules.

Supervised Learning

Learns from labels

Models the link between features and a known label, then labels new data.

Classification

The labels are discrete categories (default / no default).

Regression

The labels are continuous quantities (a loss amount).

Unsupervised Learning

No labels

Models the data with no target - 'letting the data speak for itself'.

Clustering

Identify distinct groups within the data.

Dimensionality reduction

Find a more compact representation.

Supervised Learning Algorithms

- Logistic Regression
- Linear Regression
- Ridge Regression
- Lasso Regression
- Support Vector Machines
- Trees & Random Forest
- Neural Networks

Unsupervised Learning Algorithms

K-Means

DBSCAN

Hierarchical (HCA)

PCA

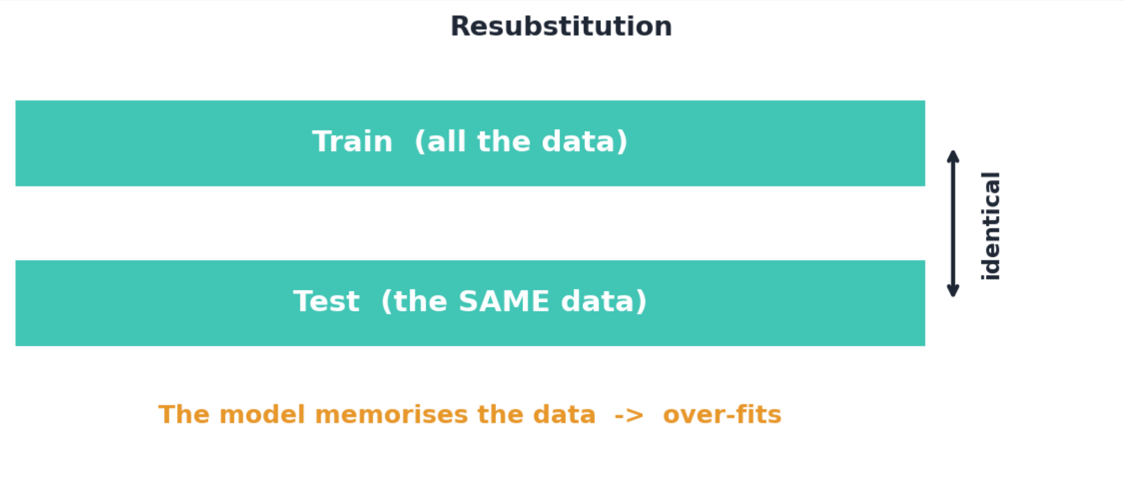
Mixture Models

Apriori

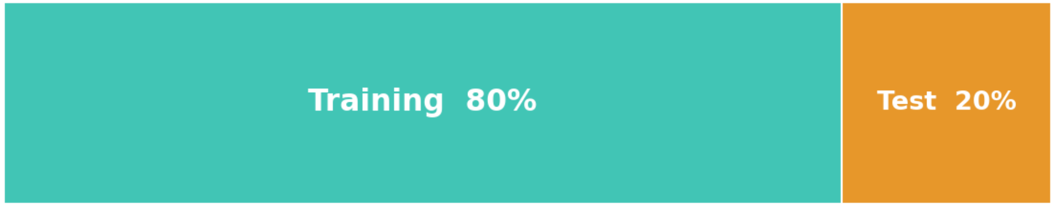
[Source: scikit-learn - Gaussian mixture models]

Model Evaluation: Resubstitution

- Train and test the model on the SAME dataset.
- The simplest possible evaluation.
- The model just memorises the data.
- That is over-fitting: great on seen data, poor on new data.



Hold-Out



1. Hold aside a subset of the data as a test set.
2. Train the model on the remaining data.
3. For example, 80% training and 20% test.
4. Measures how well the model predicts NEW data.

Cross Validation

A lucky test set?

One split might just happen to be easy or hard.

Shuffle & split

Repeat the train/test split many different ways.

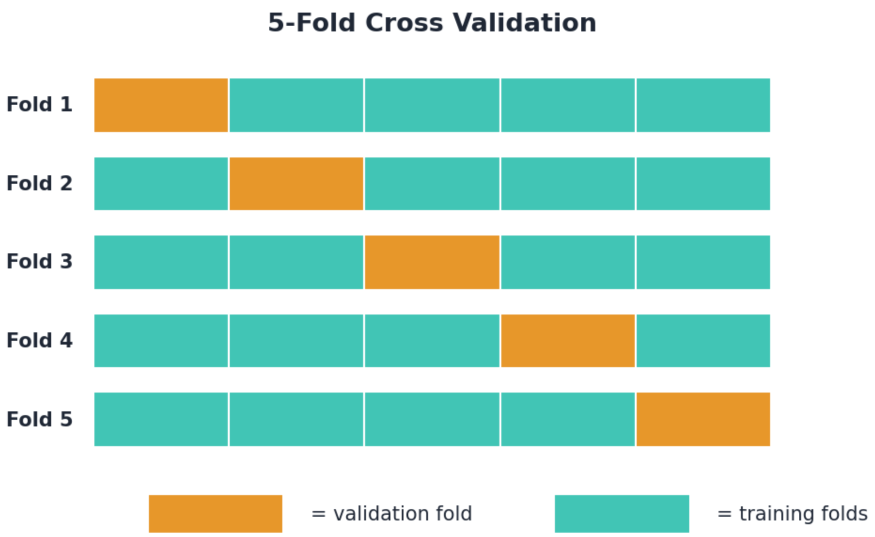
k-fold

Partition the data into k folds and rotate the test fold.

Average scores

Combine scores across folds for a reliable estimate.

Cross Validation: k-Fold



Create several partitions of the data, not just one.

Each fold takes a turn as the validation set.

Train and score the model on every split.

Average the scores for an overall measure of performance.

Evaluation Metrics: Clustering

Sum of Squared Errors

Within and between clusters.

AIC

Akaike Information Criterion.

BIC

Bayesian Information Criterion.

Evaluation Metrics: Regression

R-squared

Share of variance explained.

MSE

Mean squared error.

RMSE

Root mean squared error.

Evaluation Metrics: Classification

Accuracy

Share of correct predictions.

Precision

Of the positives flagged, how many are right.

Recall

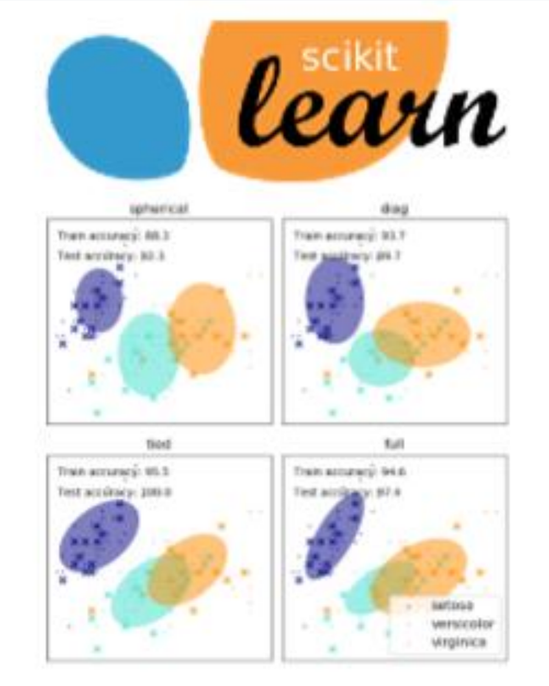
Of the real positives, how many we caught.

F1 Score

The balance of precision and recall.

Scikit-learn

- A Python machine-learning library, bundled with Anaconda.
- Create a model object for the estimator you want.
- Fit the model to some data.
- Make predictions or test the model - docs at scikit-learn.org.



[Source: scikit-learn.org](https://scikit-learn.org)

What's Next in this Session

Data Pre-processing

Preparing data for modelling.

Introduction to Regression

Predicting continuous values.

Simple & Multiple Linear Regression

Guided walkthroughs.

In this lesson, we explored

How DS is different

vs engineering and analytics.

Main tools

The data scientist's toolkit.

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Digging Deeper?

Spurious Correlations

tylervigen.com

Data Science Tutorial

GeeksforGeeks

Machine Learning Cheat Sheet

DataCamp

Questions?